

Lecture 01: Introduction to C++

C++ ek powerful aur fast programming language hai jo system-level programming, game development, aur high-performance applications ke liye use hoti hai. Is lecture mein hum C++ ke basics, binary number system, variables, data types, aur strings ko detail mein cover karenge.



What is C++

Simple Definition

C++ ek **programming language** hai jo computer programs likhne ke liye use hoti hai. Ye ek aisi language hai jo humans likh sakte hain aur computer samajh sakta hai. Iska use software, games, operating systems, aur various applications banane ke liye hota hai.

C++ mein aap instructions likhte hain jo computer ko batate hain ki kya karna hai. Ye language **high-level** bhi hai aur **low-level** control bhi deti hai, jo isse unique banata hai.

Created By

C++ ko **Bjarne Stroustrup** ne 1979 mein Bell Labs mein develop kiya tha. Is language ko originally "**C with Classes**" ke naam se jana jata tha, kyunki ye C language ke upar build ki gayi thi aur isme Object-Oriented Programming features add kiye gaye the.

Key Insight: C++ ka main goal tha C language ki speed aur power ko maintain karte hue **Object-Oriented Programming** capabilities provide karna, jo large-scale software development ko easy banati hai.

Core Philosophy

C++ ka design philosophy do cheezon par based hai: **Zero Overhead Principle** aur **Direct Mapping to Hardware**. Matlab jo features use nahi ho rahe unka koi cost nahi hota, aur jo code likhte hain wo directly hardware se connect hota hai.



Why Do We Need C++

Speed and Performance

C++ programs bahut fast run karte hain kyunki ye **compiled language** hai. Code directly machine code mein translate hota hai, jo CPU directly execute kar sakta hai. Isliye C++ ka use un jagah hota hai jahan speed critical hai.

BEST PRACTICE

Jahan microseconds matter karte hain, wahan C++ first choice hoti hai. High-Frequency Trading (HFT) systems, real-time game engines, aur operating systems isi reason se C++ mein likhe jate hain.

Control Over Computer

C++ programmer ko **memory** , **hardware** , aur **CPU** par direct control deta hai. Aap manually memory allocate aur deallocate kar sakte hain, pointers use kar sakte hain, aur hardware-specific optimizations kar sakte hain.

Real-World Applications

Field	Examples	Why C++
Games	Unreal Engine, Unity (core), GTA, Call of Duty, Fortnite	Real-time rendering, physics calculations
Systems	Windows, macOS, Linux kernels	Direct hardware access, low-level control
Apps	Photoshop, Microsoft Office, AutoCAD	Performance-critical operations
Databases	MySQL, MongoDB, Redis	Fast I/O operations, memory management
Browsers	Chrome (V8 engine), Firefox	JavaScript compilation, rendering engines
Embedded	Car systems, Drones, IoT devices	Resource constraints, real-time response
Finance	Trading systems, risk analysis	Microsecond latency requirements
ML Libraries	TensorFlow (core), PyTorch (C++ backend)	Heavy mathematical computations

Foundation for Learning

C++ seekhne se aapko **memory management** , **pointers** , aur **computer architecture** ki deep understanding milti hai. Ye knowledge aapko Java, Python, ya JavaScript jaise languages mein bhi help karti hai, kyunki aapko internally pata hota hai ki kya ho raha hai.

INTERVIEW TIP

Interviewers C++ ko "mother of all languages" mante hain. Agar aap C++ mein strong hain, to aapko memory management, pointers, aur OOP concepts ke questions confidently answer karne chahiye. Ye aapko other languages ke against edge deta hai.



Binary Number System

What is Binary

Binary ek number system hai jisme sirf **do digits** hote hain: **0** aur **1** . Ye computer ka native language hai. Computer ki puri processing 0 aur 1 par based hoti hai.

Binary mein:

- **0** = OFF (no electricity, low voltage)
- **1** = ON (electricity flowing, high voltage)

Why Binary

Computers electronic devices hain jo electricity par chalte hain. Electricity ke do stable states hote hain: **ON** aur **OFF** . Isliye binary system naturally fit hota hai. 0 aur 1 ko distinguish karna electrically easy hai, jabki 10 different voltage levels maintain karna difficult aur error-prone hota.

INTUITION

Socho ek light switch ke baare mein: ya to light ON hai ya OFF. Ye do states hain. Computer mein billions of such switches hote hain (transistors), jo milke complex calculations karte hain. Har switch ek binary digit (bit) represent karta hai.

Decimal vs Binary

Decimal	Binary	Explanation
0	0	Zero
1	1	One
2	10	One two and zero ones
3	11	One two and one one
4	100	One four, zero twos, zero ones
5	101	One four, zero twos, one one
10	1010	One eight, zero four, one two, zero one
15	1111	One eight, one four, one two, one one



Decimal to Binary Conversion

Method: Divide by 2

Decimal number ko binary mein convert karne ke liye **Division by 2** method use karte hain.

Steps:

1. Number ko 2 se divide karo
2. Remainder note karo
3. Quotient ko fir 2 se divide karo
4. Repeat jab tak quotient 0 na ho jaye
5. Remainders ko **bottom se top** padho

Example 1: Convert 13 to Binary

```
Step 1: 13 ÷ 2 = 6, remainder = 1 ↓
Step 2: 6 ÷ 2 = 3, remainder = 0 ↓
Step 3: 3 ÷ 2 = 1, remainder = 1 ↓
Step 4: 1 ÷ 2 = 0, remainder = 1 ↓
```

Read from bottom to top: 1101

Answer: **13 = 1101** in binary

Example 2: Convert 25 to Binary

Step 1: 25 ÷ 2 = 12, remainder = 1 ↓

Step 2: 12 ÷ 2 = 6, remainder = 0 ↓

Step 3: 6 ÷ 2 = 3, remainder = 0 ↓

Step 4: 3 ÷ 2 = 1, remainder = 1 ↓

Step 5: 1 ÷ 2 = 0, remainder = 1 ↓

Read from bottom to top: 11001

Answer: 25 = 11001 in binary

Example 3: Convert 8 to Binary

Step 1: 8 ÷ 2 = 4, remainder = 0 ↓

Step 2: 4 ÷ 2 = 2, remainder = 0 ↓

Step 3: 2 ÷ 2 = 1, remainder = 0 ↓

Step 4: 1 ÷ 2 = 0, remainder = 1 ↓

Read from bottom to top: 1000

Answer: 8 = 1000 in binary

Quick Practice

Decimal	Binary
5	101
7	111
10	1010
15	1111
20	10100



Binary to Decimal Conversion

Method: Powers of 2

Binary number ko decimal mein convert karne ke liye Powers of 2 method use karte hain. Steps:

- 1. Right se left jaate hue powers of 2 likho ($2^0, 2^1, 2^2, \dots$)
- 2. Har binary digit ko uski corresponding power se multiply karo
- 3. Sab results ko add karo

Powers of 2 Reference Table

Position (Right to Left)	Power	Value
0	2^0	1
1	2^1	2
2	2^2	4
3	2^3	8
4	2^4	16
5	2^5	32
6	2^6	64
7	2^7	128

Example 1: Convert 1101 to Decimal

Binary:

1

1

0

1

Power:

2^3

2^2

2^1

2^0

Value:

8

4

2

1

Step 1:

$1 \times 8 = 8$

Step 2:

$1 \times 4 = 4$

Step 3:

$0 \times 2 = 0$

Step 4:

$1 \times 1 = 1$

Total:

$8 + 4 + 0 + 1 = 13$

Answer: **1101 = 13** in decimal

Example 2: Convert 1010 to Decimal

Binary:	1	0	1	0
Power:	2^3	2^2	2^1	2^0
Value:	8	4	2	1

Step 1: $1 \times 8 = 8$

Step 2: $0 \times 4 = 0$

Step 3: $1 \times 2 = 2$

Step 4: $0 \times 1 = 0$

Total: $8 + 0 + 2 + 0 = 10$

Answer: **1010 = 10** in decimal

Example 3: Convert 11001 to Decimal

Binary:	1	1	0	0	1
Power:	2^4	2^3	2^2	2^1	2^0
Value:	16	8	4	2	1

Step 1: $1 \times 16 = 16$

Step 2: $1 \times 8 = 8$

Step 3: $0 \times 4 = 0$

Step 4: $0 \times 2 = 0$

Step 5: $1 \times 1 = 1$

Total: $16 + 8 + 0 + 0 + 1 = 25$

Answer: **11001 = 25** in decimal

Example 4: Convert 1111 to Decimal

Binary:	1	1	1	1
Power:	2^3	2^2	2^1	2^0
Value:	8	4	2	1

Step 1: $1 \times 8 = 8$

Step 2: $1 \times 4 = 4$

Step 3: $1 \times 2 = 2$

Step 4: $1 \times 1 = 1$

Total: $8 + 4 + 2 + 1 = 15$

Answer: **1111 = 15** in decimal

Quick Practice

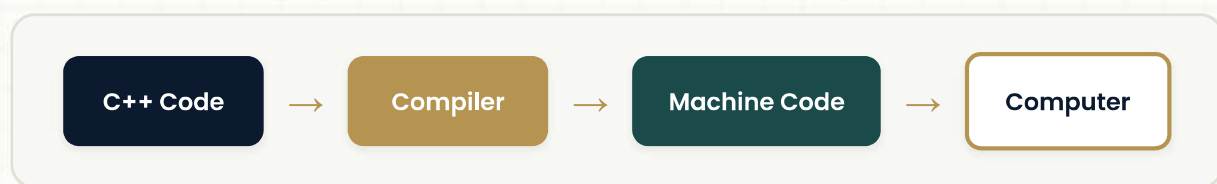
Binary	Decimal
101	5
111	7
1000	8
10100	20
11111	31



C++ Program Structure

How C++ Code Runs

C++ code directly run nahi hota. Usse pehle **compiler** usse **machine code** mein convert karta hai. Machine code wo language hai jo computer directly samajh sakta hai.



The main() Function

Har C++ program ka entry point **main()** function hota hai. Program execution yahan se start hota hai. **return 0;** ka matlab hai ki program successfully complete hua hai.

```
int main()
{
    int a = 10;

    return 0;
}
```

IMPORTANT

Kya ye program bina header file ke run hoga? **Ha, run hoga.** Sirf warning aa sakti hai.

return 0; ka matlab hai program successfully execute ho gaya hai. Compiler ke liye ye ek valid program hai, lekin production code mein hamesha proper headers include karni chahiye.

Header Files and Libraries

`cin` aur `cout` C++ language ka core part nahi hain. Ye `iostream` library se aate hain.

`#include <iostream>` ek preprocessor directive hai jo `iostream` library ko program mein include karta hai.

C++ mein 1000+ libraries hain. Jitni libraries include karoge, utna program ka size increase hoga. Isliye sirf wo libraries include karo jo actually use ho rahi hain.

```
// iostream library include karna zaroori hai
// cin, cout, endl isme defined hain
#include <iostream>

int main() {
    int a = 10;
    std::cout << a;
    return 0;
}
```

Namespace std

`std::cout` mein `std` ka matlab `Standard` hai aur `::` `Scope Resolution Operator` hai. Ye batata hai ki `cout` standard namespace mein defined hai.

Bar bar `std::` likhne se bachne ke liye **using namespace** `std`; use karte hain. Isse aap directly `cout`, `cin`, `endl` use kar sakte hain.

```
#include <iostream>
using namespace std;

int main() {
    int a = 10;
    cout << a << endl;
    return 0;
}
```

EDGE CASE

using namespace std; small programs mein convenient hai, lekin large projects mein naming conflicts ho sakte hain. Professional code mein specific identifiers import karna better practice hai: **using** `std::cout`; **using** `std::endl`;



Variables

What is a Variable

Variable memory mein ek container hota hai jo data store karta hai. Isme aap value daal sakte hain, change kar sakte hain, aur use kar sakte hain.

```
int age = 25;
```

Is example mein:

- **int** = container ka type (integer)
- **age** = container ka naam
- **25** = stored value

Rules for Naming Variables

Allowed	Not Allowed	Reason
<code>int age;</code>	<code>int 123roll;</code>	Number se start nahi kar sakte
<code>int student_name;</code>	<code>int student-name;</code>	Hyphen allowed nahi hai
<code>int rollNo123;</code>	<code>int int;</code>	Keywords use nahi kar sakte
<code>int _value;</code>	<code>int my name;</code>	Spaces allowed nahi hain

Declaring Variables

C++ mein variables declare karne ke 3 tareeke hain:

METHOD 1: DECLARE THEN ASSIGN

```
int age;  
age = 25;
```

METHOD 2: DECLARE AND ASSIGN TOGETHER

```
int age = 25;
```

METHOD 3: MULTIPLE VARIABLES

```
int a = 5, b = 10, c = 15;
```

BEST PRACTICE

Hamesha variables initialize karo jab declare karte ho. Uninitialized variables mein **garbage values** hoti hain jo unpredictable behavior cause kar sakti hain. `int x; // Avoid` - contains garbage value `int x = 0; // Good - explicit initialization`



Data Types

Basic Data Types

Data Type	Used For	Size	Example
int	Whole numbers	4 bytes	int age = 25;
double	Decimal numbers	8 bytes	double price = 99.99;
char	Single character	1 byte	char grade = 'A';
bool	True/False	1 byte	bool isPassed = true;
string	Text	Varies	string name = "Rohit";

Complete Example

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    int age = 25;
    double height = 5.9;
    char grade = 'A';
    bool isStudent = true;
    string name = "Rohit";

    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Height: " << height << endl;
    cout << "Grade: " << grade << endl;
    cout << "Student: " << isStudent << endl;

    return 0;
}
```

Important Notes

IMPORTANT

char ke liye hamesha **single quotes** use karo: 'A'. Double quotes "A" string literal banate hain, jo char type se different hai.

string ke liye hamesha **double quotes** use karo: "Rohit". Single quotes 'Rohit' compile error denge kyunki single quotes sirf single character ke liye hain.



Strings

Basic String Operations

C++ mein **string** ek sequence of characters hai. **#include <string>** library include karni padti hai string operations ke liye.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string name = "Rohit";

    // Length of string
    cout << name.length() << endl; // Output: 5

    // Combine strings (Concatenation)
    string firstName = "Rohit";
    string lastName = "Kumar";
    string fullName = firstName + " " + lastName;
    cout << fullName << endl; // Output: Rohit Kumar

    // Access characters (0-based indexing)
    cout << name[0] << endl; // Output: R
    cout << name[1] << endl; // Output: o

    return 0;
}
```

String Input with Spaces

cin sirf ek word read karta hai. Agar input mein spaces hain, to wo space ke baad ka part ignore kar deta hai.

EDGE CASE

Problem: `string` name; cin >> name; *// Input: "Rohit Kumar" // Stores: "Rohit" only*

>> operator whitespace (space, tab, newline) ko delimiter manta hai. Isliye space ke baad ka data buffer mein reh jata hai aur next input mein chala jata hai.

Solution: `getline()` function use karo jo entire line read karta hai spaces ke saath.

```
string name;
getline(cin, name); // Reads entire line including spaces
// Input: "Rohit Kumar"
// Stores: "Rohit Kumar"
```

BEST PRACTICE

Mixed input scenarios mein (cin ke baad getline), buffer clear karna important hai. `int` n; cin >> n; cin.ignore(); `string` s; getline(cin, s); *// cin.ignore() newline character ko discard karta hai*



Key Takeaways

TAKEAWAY

- C++ ek fast, powerful, compiled language hai jo direct hardware control deti hai
- Computer sirf binary (0 aur 1) samajhta hai — decimal to binary divide-by-2 se, binary to decimal powers-of-2 se convert hota hai
- Har C++ program main() function se start hota hai aur return 0 successful execution indicate karta hai
- iostream library cin, cout provide karti hai; using namespace std; std:: likhne se bachata hai
- Variables memory containers hain; hamesha initialize karke declare karo
- char ke liye single quotes ('A'), string ke liye double quotes ("text") use karo
- Spaces ke saath string input ke liye getline(cin, variable) use karo

Practice Exercise

PATTERN

Exercise 1: Convert following decimal numbers to binary: 17, 31, 42, 100.

Exercise 2: Convert following binary numbers to decimal: 1001, 10101, 111000, 100000.

Exercise 3: Write a C++ program jo user se name, age, aur height input le aur formatted output print kare. Height input decimal mein honi chahiye. Spaces ke saath name handle karo.

Exercise 4: Bina using namespace std; ke same program likho. Har jagah std:: prefix use karo aur difference observe karo.

INTERVIEW TIP

Interview mein commonly puche jane wale questions:

1. C++ aur C mein kya difference hai? (OOP features, classes, inheritance, polymorphism)
2. Compiled vs Interpreted language kya hoti hai?
3. Binary conversion ka real-world use kya hai? (Memory addresses, flags, bitwise operations)
4. Variable initialization kyun important hai? (Garbage values, undefined behavior)
5. getline aur cin mein kya difference hai? (Delimiter handling, buffer behavior)